

★ WEEK 13 · PART 3 of 3

Intervals, Prediction, and Why Bayes Suits Finance

Credible Intervals, the Posterior Predictive, and the Catch

Credible vs confidence · posterior predictive · the computing problem

A posterior does more than name a value: it gives credible intervals and honest forecasts. We contrast it with the frequentist interval, show why finance loves it, and meet the catch that MCMC will solve.

3

PART 3 · INTERVALS & PREDICTION

From Posterior to Decisions

Credible versus confidence intervals, the posterior predictive for honest forecasts, why Bayesian methods fit finance, and the computational catch that motivates Markov chain Monte Carlo.

From a Posterior to Answers

Parts 1-2 built the posterior; Part 3 uses it

1

We can compute a posterior

A full distribution over the parameter, from prior and data. Now: what do we actually DO with it?

2

It answers real questions

Credible intervals for uncertainty, and the posterior predictive for forecasts — both fall straight out of the posterior.

3

And it fits finance — with a catch

Short, noisy data make honest error bars essential. But computing posteriors for real models is hard, which sets up next week.

Part 3 in one line: read intervals and forecasts off the posterior — then confront the computing problem.

Credible vs Confidence Intervals

They are not the same thing

A credible interval and a confidence interval sound alike but mean different things. The credible interval is a probability statement about the PARAMETER; the confidence interval is a statement about the PROCEDURE across repeated samples.

Credible (Bayesian)

- 'There is a 95% probability θ is in this range.'
- A direct statement about the parameter.
- Read straight off the posterior.

Confidence (frequentist)

- '95% of such intervals would contain θ .'
- A statement about the method, not this θ .
- The parameter is fixed; the interval is random.

The Credible Interval

A direct probability about the parameter

credible interval: $P(L \leq \theta \leq U \mid \text{data}) = 0.95$

What it says

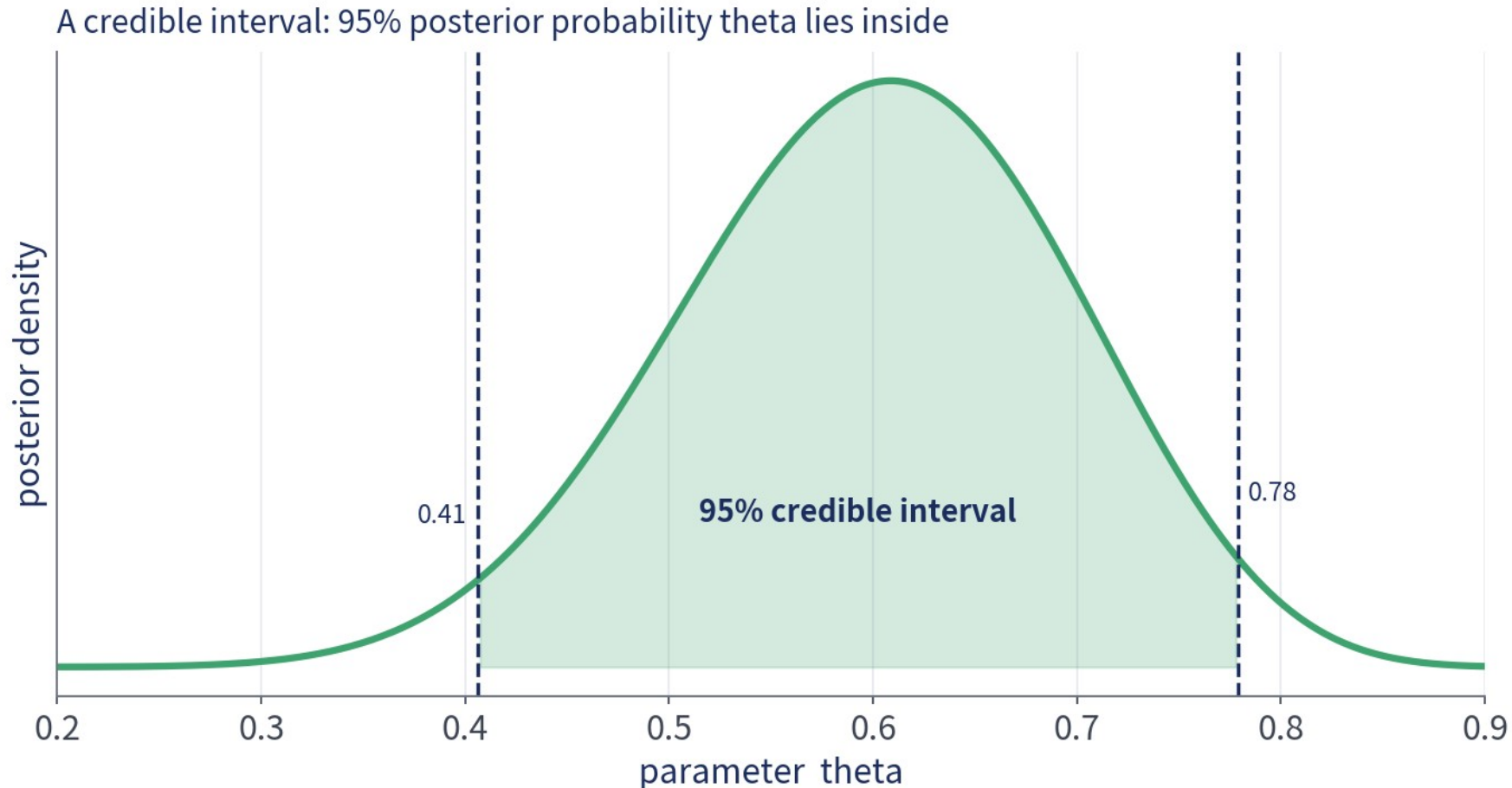
- The parameter lies in the range with 95% probability.
- That probability is given THIS data.
- It is exactly the intuitive reading people want.

How you get it

- Take the region holding 95% of the posterior.
- Equal-tailed, or the highest-density region.
- No repeated-sampling story needed.

A Credible Interval

Shade the central 95% of the posterior and you have the credible interval. It means exactly what it sounds like: 95% posterior probability that the parameter lies inside the shaded band.



The Confidence Interval

A statement about the procedure

What it actually means

- Imagine repeating the study many times.
- 95% of the intervals produced would cover θ .
- It is a property of the method, over repetitions.

What it does NOT mean

- It is NOT a 95% probability for THIS interval.
- The parameter is fixed; it is in or out.
- The common intuitive reading is technically wrong.

Two Kinds of Interval

The credible interval makes probability a statement about the parameter, given this data. The confidence interval makes it a statement about the procedure, over hypothetical repeated samples.

Two Kinds of Interval

They sound alike but say very different things - one about the parameter, one about the procedure.

Credible interval (Bayesian)

"Given THIS data, there is a 95% probability that θ lies in $[L, U]$."

- A statement about the PARAMETER.
- Uses this one dataset.
- Read straight off the posterior.
- Says what people usually MEAN.

Probability is about θ .

Confidence interval (frequentist)

"95% of such intervals, over many repeated samples, contain θ ."

- A statement about the PROCEDURE.
- Imagines many repeated samples.
- θ is fixed; the interval is random.
- NOT a probability about this θ .

Probability is about the method.

Why the Difference Matters

The interpretation you want

The practical upshot

- People almost always WANT the credible reading.
- The Bayesian interval delivers it directly.
- The frequentist interval is routinely misread as it.

When they roughly agree

- With lots of data and a weak prior, they nearly coincide.
- The gap is widest with small samples or strong priors.
- The MEANING differs even when the numbers are close.

The Posterior Predictive

Forecasting the Bayesian way

To forecast, a Bayesian doesn't just plug in a best parameter. The posterior predictive averages the forecast over the ENTIRE posterior, folding parameter uncertainty into the prediction itself.

The idea

- For each possible parameter, form a forecast.
- Weight each forecast by its posterior probability.
- Average them into one predictive distribution.

Why it is better

- It admits we don't know the parameter exactly.
- So the forecast is honestly wider.
- It avoids the false precision of a plug-in.

The Posterior Predictive, Formally

Integrate the forecast over the posterior

$$\text{posterior predictive: } p(y_{\text{new}} \mid \text{data}) = \int p(y_{\text{new}} \mid \theta) p(\theta \mid \text{data}) d\theta$$

Reading it

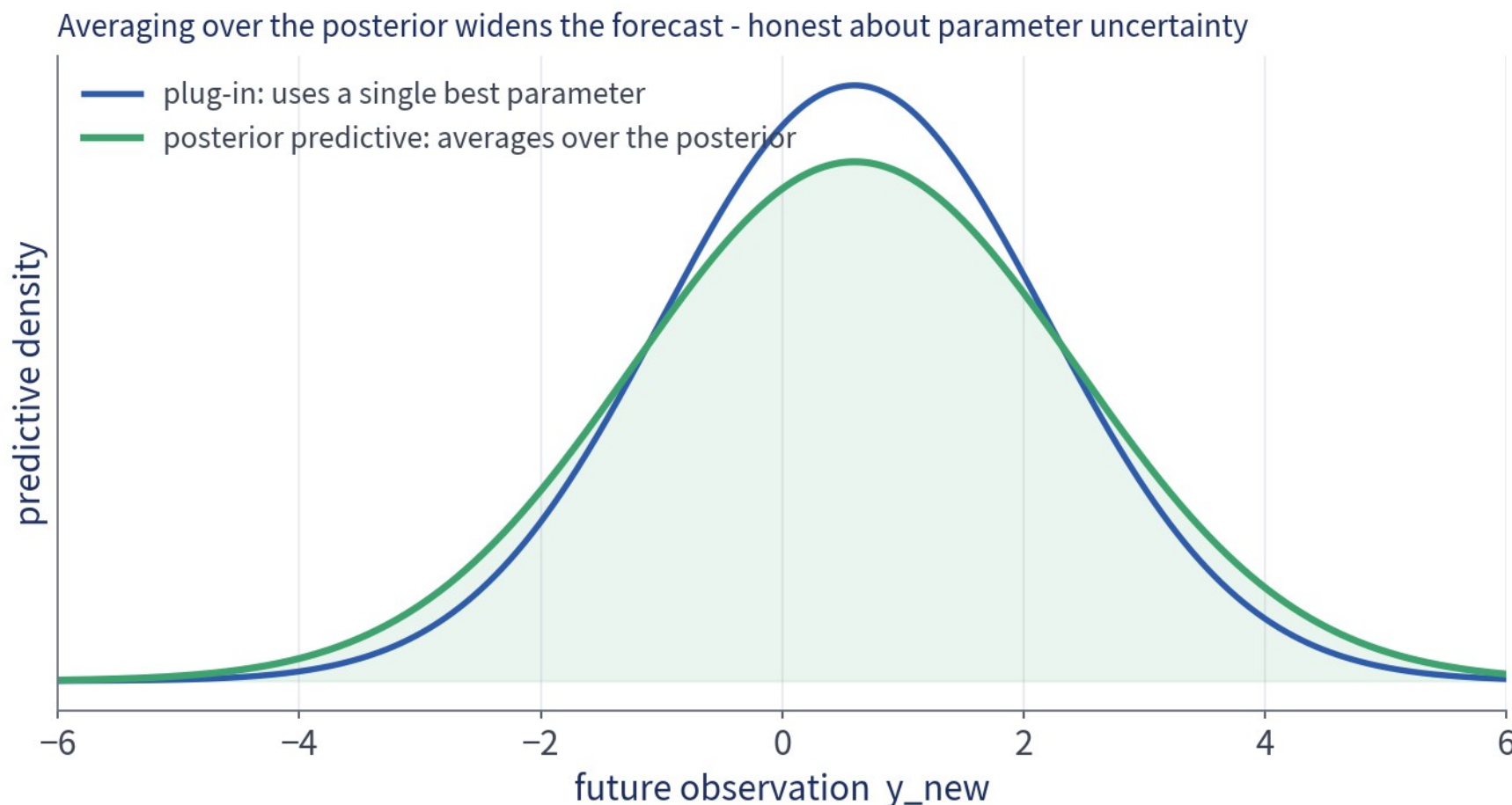
- Inside: the forecast for a fixed parameter.
- Weighted by: the posterior for that parameter.
- Integrated over: all possible parameter values.

The result

- A single predictive distribution for new data.
- Wider than any single-parameter forecast.
- It carries both data noise and parameter uncertainty.

Averaging Widens the Forecast

A plug-in forecast (blue) uses one best parameter and looks confident. The posterior predictive (green) averages over the posterior — wider, with heavier tails, and honest about what we don't know.



Point Forecast vs Predictive Distribution

A number versus a distribution of outcomes

A point forecast

- A single predicted value for the future.
- It hides how uncertain that prediction is.
- It can look far more precise than it is.

A predictive distribution

- A full distribution over future outcomes.
- Its spread reflects true forecast uncertainty.
- You can read off any prediction interval you need.

Why Bayesian Methods Suit Finance

Where the approach earns its keep

Finance is the natural home for Bayes: data is short and noisy, regimes shift, and analysts hold real prior views. Bayesian methods turn all three from problems into features — honest uncertainty, stabilized estimates, and a place for judgment.

The conditions

- Short samples and low signal-to-noise.
- Structural change — the past isn't fully the future.
- Analysts with genuine prior views.

The Bayesian answer

- Honest error bars instead of false precision.
- Priors that stabilize small-sample estimates.
- A principled slot for informed judgment.

Small Samples and Structural Change

Finance rarely gives you clean, abundant data

Why estimates are fragile

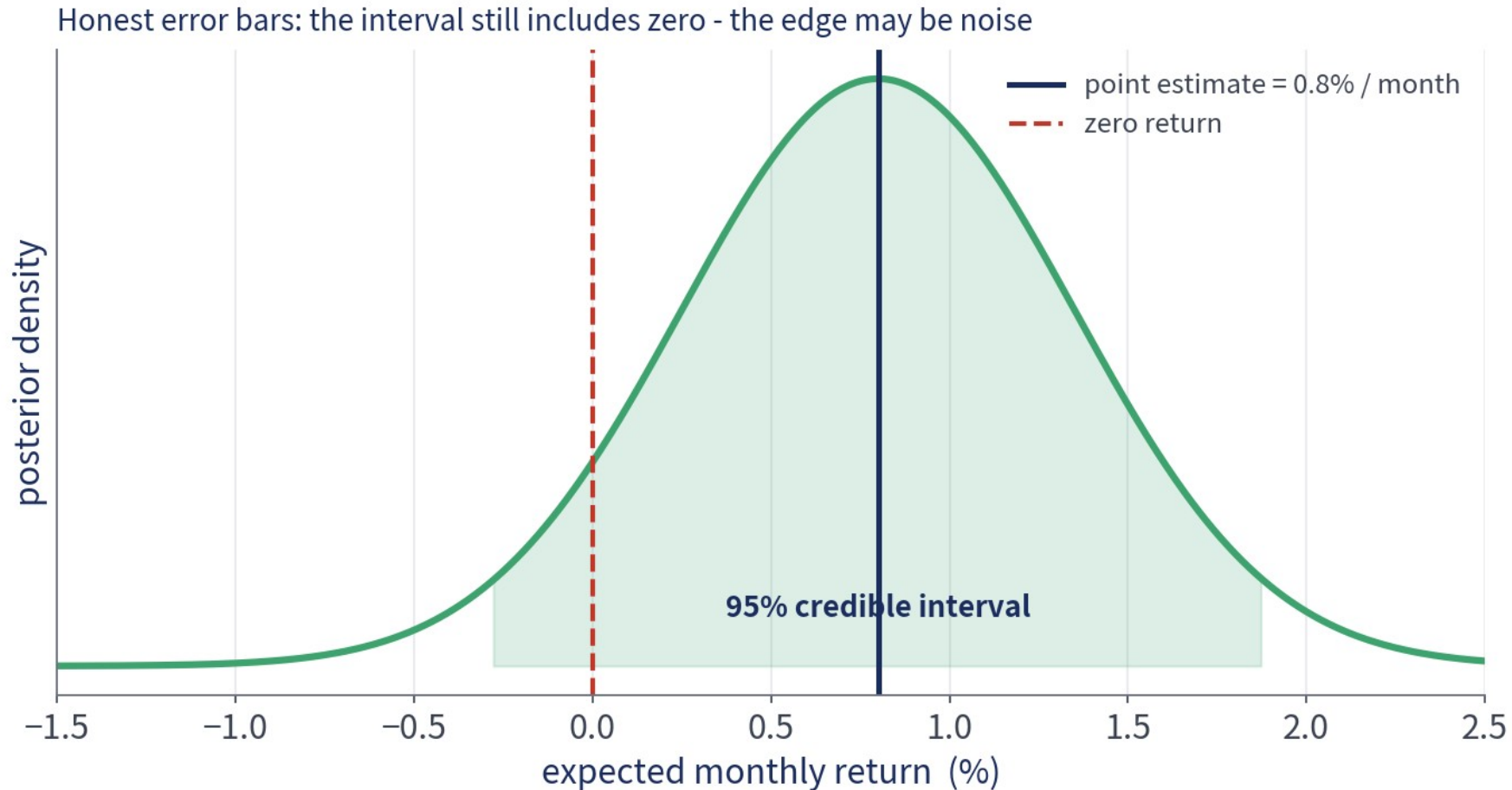
- Return series are short relative to their noise.
- Relationships drift and break over time.
- A point estimate can be badly overconfident.

How Bayes helps

- Priors shrink wild small-sample estimates.
- Posteriors show honestly how little we know.
- You size decisions to real, not assumed, certainty.

Honest Error Bars on a Return Estimate

A posterior for an expected monthly return, from a short sample. The point estimate looks positive, but the 95% credible interval still spans zero — the apparent edge may be noise. That caution is the value.



Frequentist vs Bayesian

Two valid tools, different strengths

Aspect	Frequentist	Bayesian
Parameter	a fixed unknown	a distribution
Main output	point estimate + CI	a full posterior
Interval meaning	about the procedure	about the parameter
Prior	none	explicit
Small samples	can be unstable	stabilized by the prior

Neither is 'right' — they answer different questions. Bayes shines when uncertainty must be explicit.

When to Use Which

Matching method to problem

Lean frequentist when

- Data is abundant and the model is simple.
- You want a quick, standard, defensible estimate.
- No strong prior knowledge is at stake.

Lean Bayesian when

- Data is scarce or the model is complex.
- You have real prior knowledge to include.
- Honest, propagated uncertainty is the point.

The Catch: Computing the Posterior

Why Bayes waited 200 years for computers

Everything rested on one integral — the normalizer. For the Beta-Binomial it was free, thanks to conjugacy. For real models it is intractable, and that single obstacle held Bayesian methods back for two centuries.

The easy cases

- Conjugate models update by arithmetic.
- The normalizer cancels or has a formula.
- No integral to actually compute.

The hard cases

- Realistic models have no conjugate shortcut.
- The normalizer is a high-dimensional integral.
- It cannot be done exactly, by hand or otherwise.

Why Exact Bayes Doesn't Scale

The curse of dimensions

The problem

- The normalizer integrates over every parameter.
- With many parameters, that space is enormous.
- Grid or analytic integration becomes impossible.

The consequence

- Exact posteriors exist only for special models.
- Real financial models are not so lucky.
- We need a way to work with the posterior indirectly.

The Bridge to MCMC

Sample the posterior instead of computing it

The key idea

- Don't compute the posterior — draw SAMPLES from it.
- Samples need only the shape, not the normalizer.
- From samples you get means, intervals, everything.

What makes it possible

- Markov chain Monte Carlo builds such samplers.
- It sidesteps the intractable integral entirely.
- That is exactly next week's subject.

Error Bars on Everything We Built

Every fitted number from this course — AR coefficients, GARCH volatility, a hidden state, a regime probability, rate parameters — can become a posterior. Bayes adds an honest error bar to each; MCMC makes it computable

Error Bars on Everything We Built

Every point estimate from this course can become a posterior distribution.

AR / ARMA

point estimate of the coefficients



GARCH

point estimate of the volatility



VAR

point estimate of the dynamics



Kalman

point estimate of the hidden state



HMM

point estimate of the regime probs



OU / rates

point estimate of kappa, theta, sigma



The Bayesian upgrade

Wherever we reported a single fitted number, Bayes replaces it with a posterior - a best guess AND an honest error bar. In finance, where data is short and noisy, that uncertainty is not a nicety - it is the point.

The catch: computing these posteriors needs the tools of Week 14 (MCMC).

Credible Intervals and Prediction

Direct Claude Code to read a posterior fully

GOAL From a posterior, extract a credible interval and a posterior predictive, and contrast with a plug-in.

[INPUT] Prompt · Claude Code

Write and run `posterior_use.py` that:

1. takes a posterior (e.g. Beta or Normal) from a Bayesian fit of a probability or a mean return;
2. computes and plots the 95% CREDIBLE interval;
3. builds the POSTERIOR PREDICTIVE for a new observation by averaging over the posterior;
4. overlays a PLUG-IN forecast that uses only the posterior mean, to show it is too narrow.

Report whether the credible interval includes a 'no-edge' value (like zero return).

How to work with it

- The credible interval reads off the posterior.
- The predictive is wider than the plug-in.
- An interval spanning zero is a real warning.
- This is the honest way to forecast.

Deliverable: `posterior_use.py` + a credible interval and a predictive-vs-plug-in comparison.

Three Questions Before You Finish

If you can answer these, Week 13 has landed

Q1	A credible interval is a probability statement about what?	The parameter
Q2	Why is the posterior predictive wider than a plug-in forecast?	It adds parameter uncertainty
Q3	What single obstacle held Bayes back for 200 years?	Computing the normalizer

All three come from the credible interval, the predictive average, and the intractable integral.

Three Pitfalls to Avoid

Where Bayesian practice slips

Common mistakes

- "A confidence interval is a 95% probability for this θ ."
- "A plug-in forecast captures the uncertainty."
- "A positive point estimate means a real edge."

The reality

- That reading is the CREDIBLE interval, not the confidence one.
- Only the predictive folds in parameter uncertainty.
- Check the interval — it may well span zero.

Week 13 in Three Ideas

Pulling the whole week together

1

Bayes updates belief into a posterior

Parts 1-2: posterior proportional to likelihood times prior, made concrete by the Beta-Binomial. A parameter becomes a distribution, updated by data, with conjugacy turning the update into arithmetic.

2

The posterior answers real questions

Part 3: credible intervals give a direct probability about the parameter, and the posterior predictive gives honest forecasts that carry parameter uncertainty — a natural fit for short, noisy financial data.

3

The catch is computation

Real posteriors need an intractable integral. That single obstacle is what Markov chain Monte Carlo overcomes — by sampling the posterior instead of computing it.

Next — Week 14: Markov chain Monte Carlo, the engine that makes Bayesian inference practical.